

Statistics

Statistics is the science of collecting, organizing and analyzing data.

Data:- "fact or pieces of information"

Types of Statistics:

① Descriptive statistics
→ It consists of organizing and summarizing data.

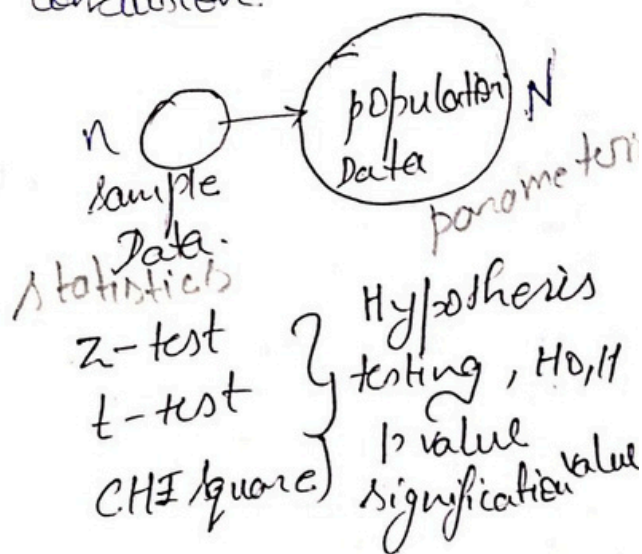
Sub Types:

* Measure of Central Tendency
[mean, median, mode]

* Measure of Dispersion
[Variance, std]

* Different type of Distribution of data.
[Histogram, pdf, pmf]

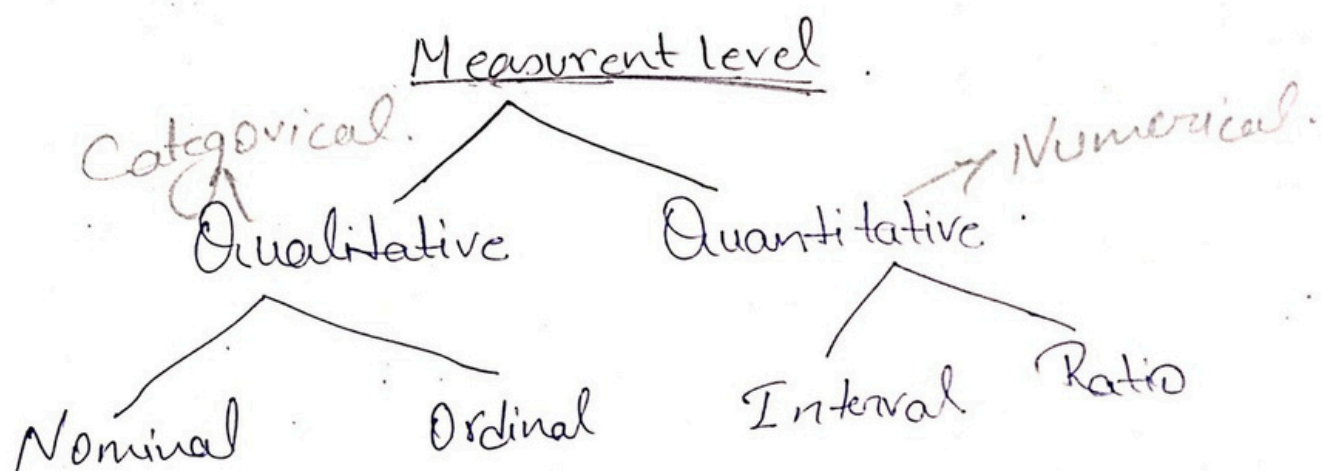
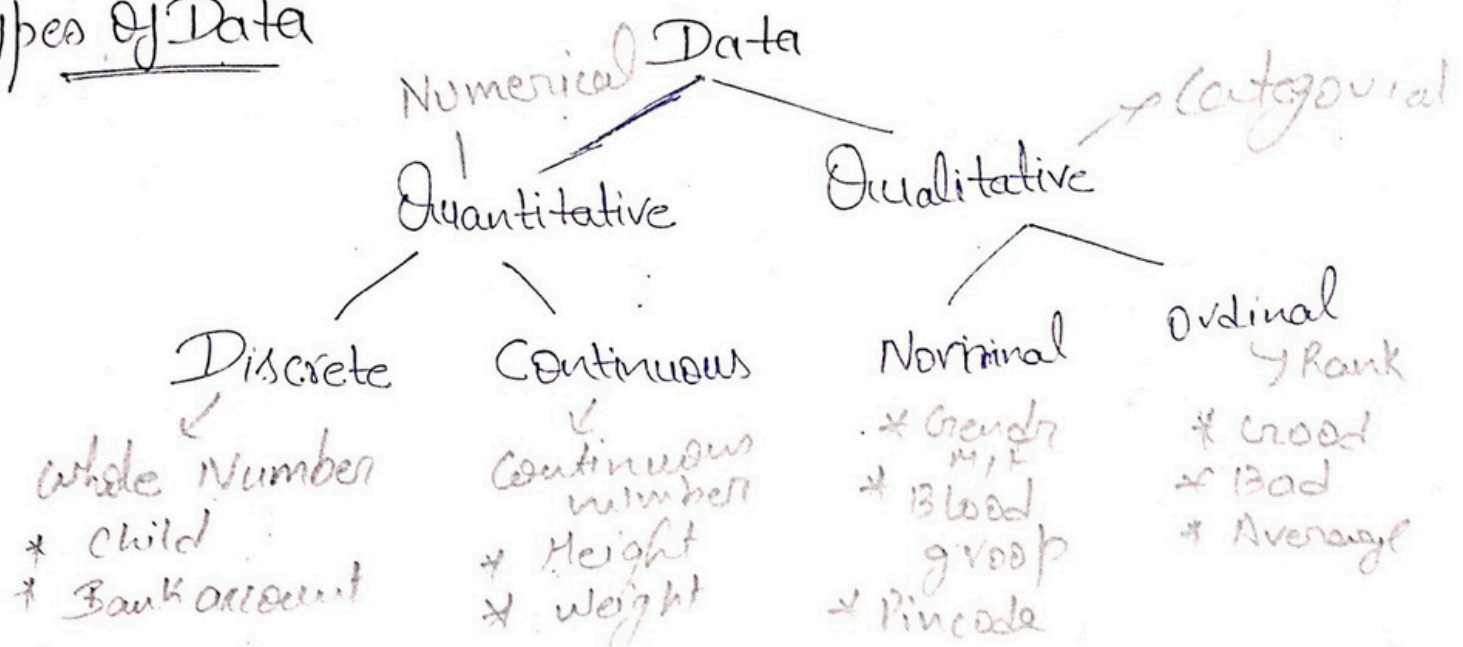
② Inferential Statistics
→ It consists of using data you have measured to form conclusion.



Population and Sample Data.

Population:- the group you are interested in studying
sample:- a subset of population.

Types of Data



* Nominal Scale Data

- * Order does not matter
- * Gender, colors, labels

eg! Favorite color

Red	→ 5	}	50%
Blue	→ 3		30%
Orange	→ 2		20%

* Ordinal Scale Data

- * Ranking is important
- * Order matter
- * Difference cannot be measured

eg! 1 → Best } Differ
2 → Good }
3 → Bad } Differ

* Interval Data (context)

- * The order matter
- * Difference can be measured
- * Ratio cannot be measured
- * No true "0" starting point

exception
OK → -273.15

eg! Temperature

30°F	}	30°C
60°F		30°C
120°F		60°C

[3:1] X

∞ ∞ +

* Ratio Scale Data (context)

0°K $\rightarrow$ -273°K

eg: student marks in class
1st mark = 90, 60, 30, 10, 0
ratio = 90:30
 $= (3:1)$ ✓

* The order matter

* Difference are measurable
(including ratio)

* Contains a "0" starting point

$\rightarrow$ 0 mark, starting
no negative

Measure of Central Tendency: (Descriptive Statistics)

* Mean, * Median, * Mode

(i) Mean:

Population (N)

$$\text{Population mean } (\mu) = \frac{\sum_{i=1}^n x_i}{N}$$

Sample (n)

$$\text{Sample mean} = \frac{\sum_{i=1}^n x_i}{n}$$

(ii) Median:

* Sort the Random variable.

* No. of element.

if count is even the $(\text{mid} + \text{mid} + 1) / 2$
else mid

* It is used to find the Central Tendency when outliers are present.

(iii) Mode:

* frequency maximum

* for categorical [1, 1, 1, 2, 3, 3, 4, 5, 5] $\rightarrow 1$

2) Measure of Dispersion [spread of the Data]

* Variance * standard deviation

i) Variance

population Variance

$$\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$

x_i = Data points

μ = population mean

N = Population size

Let!

$$S^2 = 2.5$$



sample Variance

$$S^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}$$

x_i = Data Points

$\bar{x}$ = Sample mean

n = Sample size.

(n-1) - unbiased estimator to create.

$$S^2 = 7.5$$

Variance spread.



2) Standard Deviation

Population Standard Deviation

$$\sigma = \sqrt{\text{Variance}}$$

Let! 1, 2, 3, 4.5

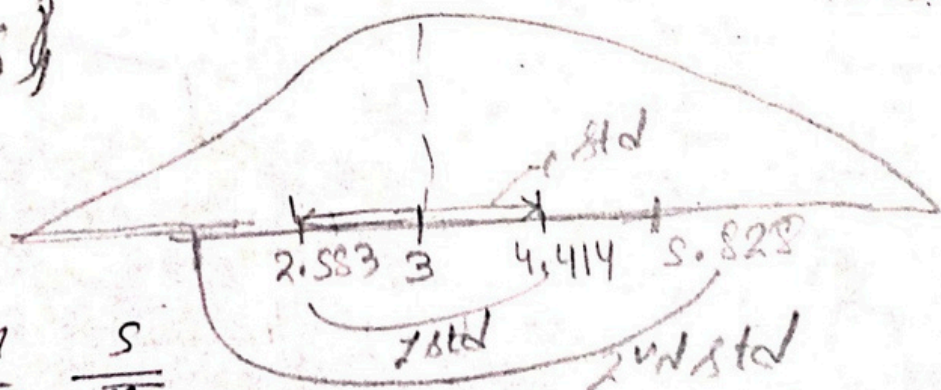
$$\bar{x} = 3$$

$$\sigma = 2.414$$

Sample Std

$$\text{Std} = \sqrt{S^2}$$

S^2 = Sample Variance



Sample ~~standard~~ deviation coefficient of variation = $\frac{S}{\bar{x}}$

Variance, Standard Deviation (SD), and Coefficient of Variation (CV) with a simple real-life analogy

🧩 Step 1: Variance — Total Spread

Variance tells you how widely your arrows are scattered overall.

- If your arrows are all over the target, the **variance is high**.
- If most arrows are close together, the **variance is low**.

💡 It measures the **overall scatter** — but since it's in squared units, it's like saying "the *total area* of scattering," which isn't easy to visualize directly.

🎯 Step 2: Standard Deviation — Average Deviation

Standard deviation gives a clearer sense — it's like saying:

“On average, each arrow lands X cm away from the bullseye.”

It's the **average miss distance**, in the **same units** as your score or distance.

So if SD = 5 cm, your arrows typically land about 5 cm away from the mean position.

⚖️ Step 3: Coefficient of Variation — Relative Consistency

Now imagine:

- You shoot from **10 meters** away.
- Your friend shoots from **100 meters** away.

You both have SD = 5 cm, but are you equally consistent? 🤔

Not really — for your friend, missing by 5 cm at 100 meters is **much more accurate** than missing 5 cm at 10 meters.

That's where CV comes in.

CV compares the spread (SD) relative to the mean (average distance or score).

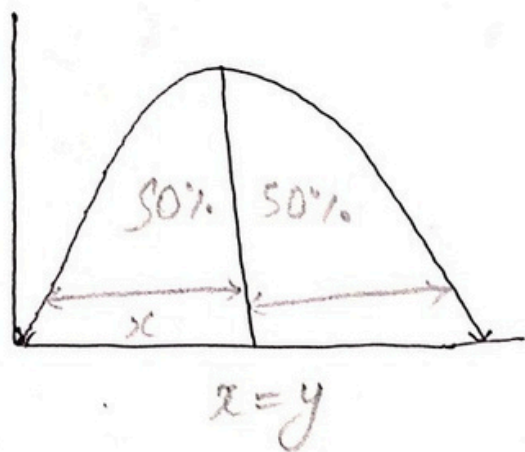
$$CV = \frac{SD}{Mean} \times 100\%$$

- Your CV (5 cm / 10 m = 0.5%) → less consistent
- Friend's CV (5 cm / 100 m = 0.05%) → more consistent

So CV says **who's more precise relative to their average performance** — regardless of scale or units.

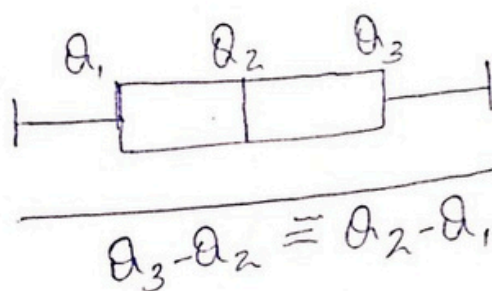
Skewness: measure of the asymmetry

* Normal / Gaussian / Symmetrical Distribution

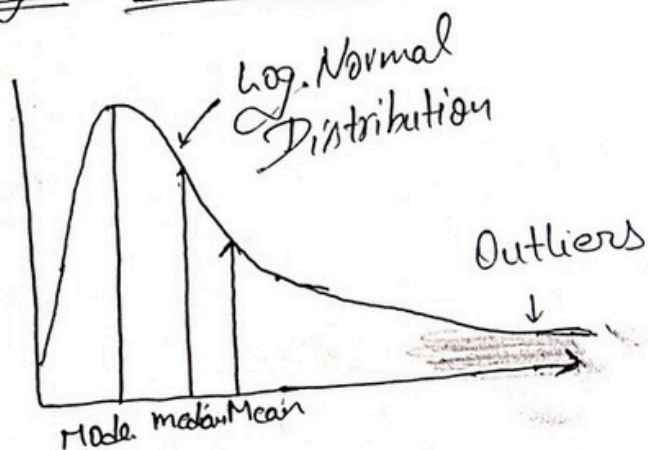


* No skewness

* Mean = Median = Mode



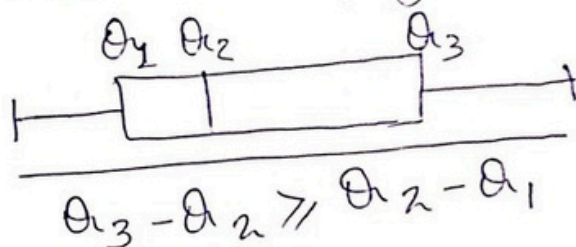
* Right Skewed



* Positive skewed

* mean > median > mode

* outliers on right side



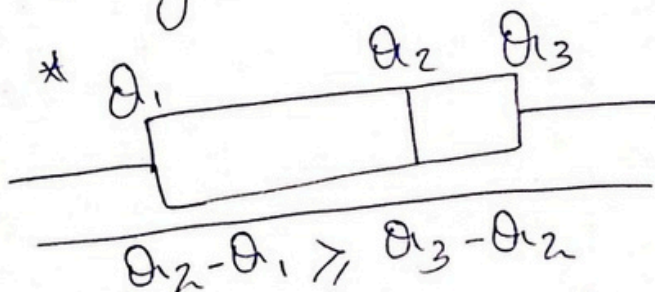
* Left Skewed Distribution



* mean < median < mode

* outliers on left side

* Negative skewed



Covariance (Direction only)

Covariance measure the direction of the linear relationship between two variable.

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{n} \rightarrow \text{Population.}$$

$$S_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1} \rightarrow \text{Sample formula}$$

* Covariance gives a sense of direction

* ^{greater than} (>0) the two variables move together

* ^{less than} (<0) the two variables move in opposite directions

* $(=0)$ the two variables are independent.

* Covariance is not standardized, so its value depends on the scale of the variable eg! $(-\infty \text{ to } \infty)$

Covrelation: ^{coefficient} Indicates direction & strength.

* Standardized (unit-free)

* -1 to $+1$

* Indicates direction & strength

* Use to compare relationships across datasets with different scales.

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Types: * Pearson's $\rightarrow$ same
* Spearman's $\rightarrow$ $\frac{\text{Cov}(R(X), R(Y))}{(R(X)) \cdot (R(Y))}$
Rank

Covrelation range $(-1 \text{ to } 1)$
* 1 perfect positive corr
* -1 perfect negative
* 0 No linear relation

σ_X, σ_Y are the standard deviation of X and Y

If you take the covariance of a variable with itself, you get its variance.

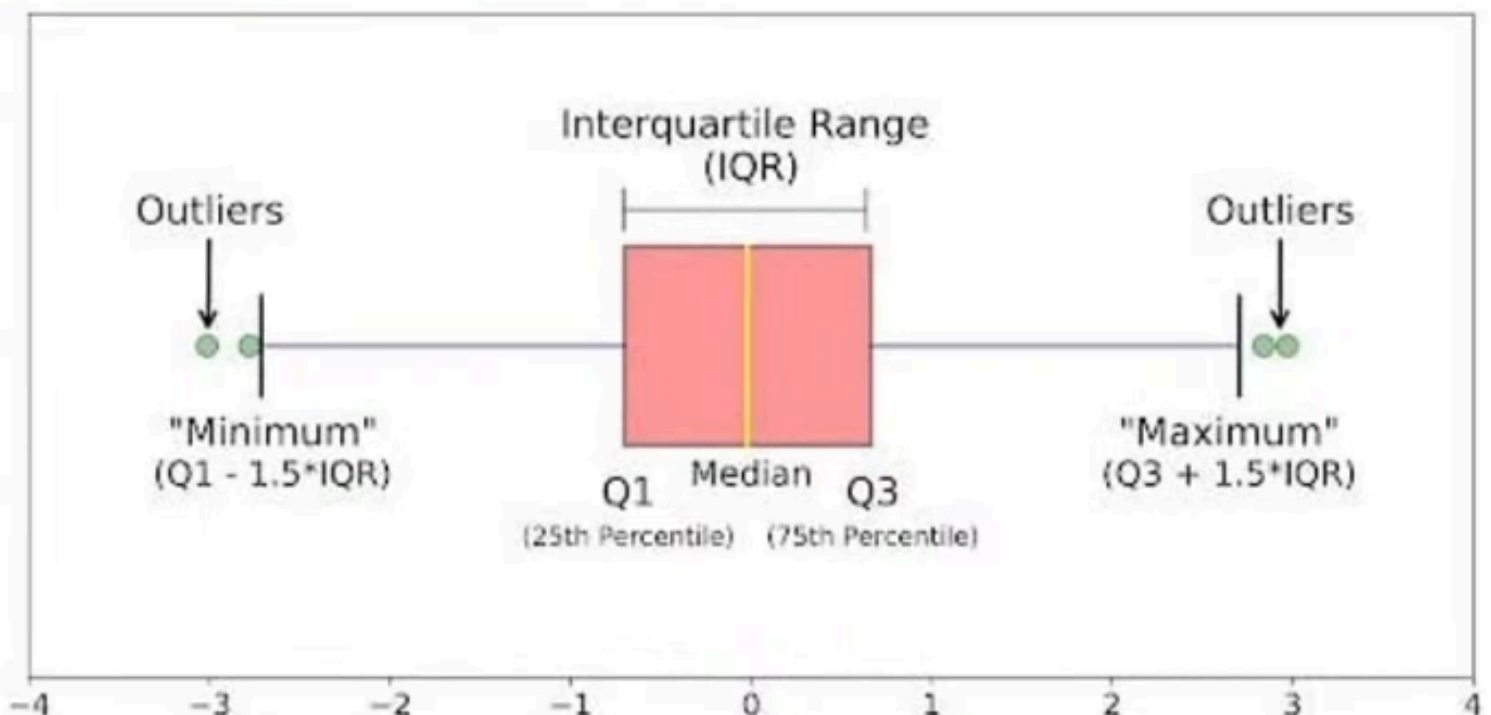
$$\text{Yes, } \text{Cov}(X, X) = \text{Var}(X)$$

“Correlation does not imply causation” means that just because two things are related or change together, it doesn't necessarily mean that one causes the other

When two variables show a correlation, they move together in a predictable pattern—when one increases, the other might increase (or decrease). However, this relationship alone doesn't prove that one variable is directly causing the change in the other.

What is a boxplot

A box plot, also known as a box-and-whisker plot, is a graphical representation of a dataset that shows the distribution of the data. The box plot displays a summary of the data, including the minimum and maximum values, the first quartile (Q1), the median (Q2), and the third quartile (Q3).

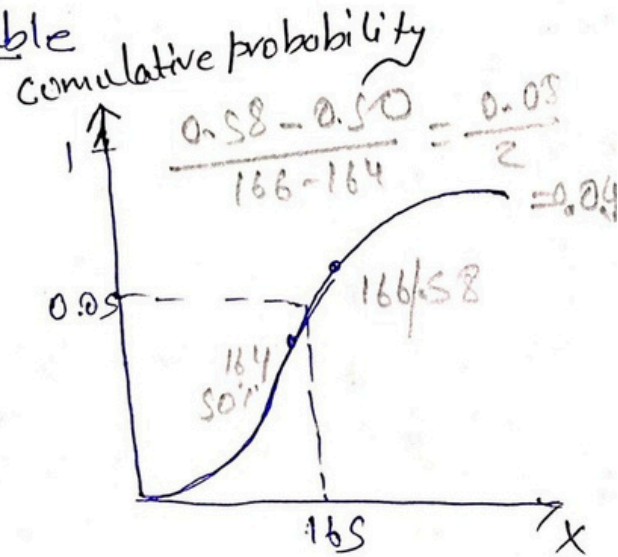
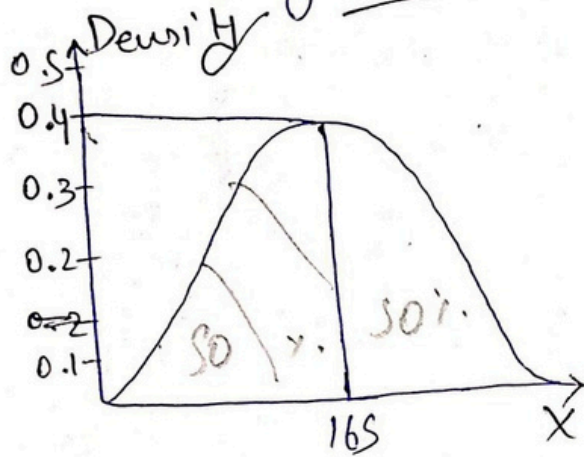


Advanced Probability Distribution function (Density function)

Pdf / Pmf / CDF / Cumulative Distribution function

* Probability Density function (Pdf) / (CDF)

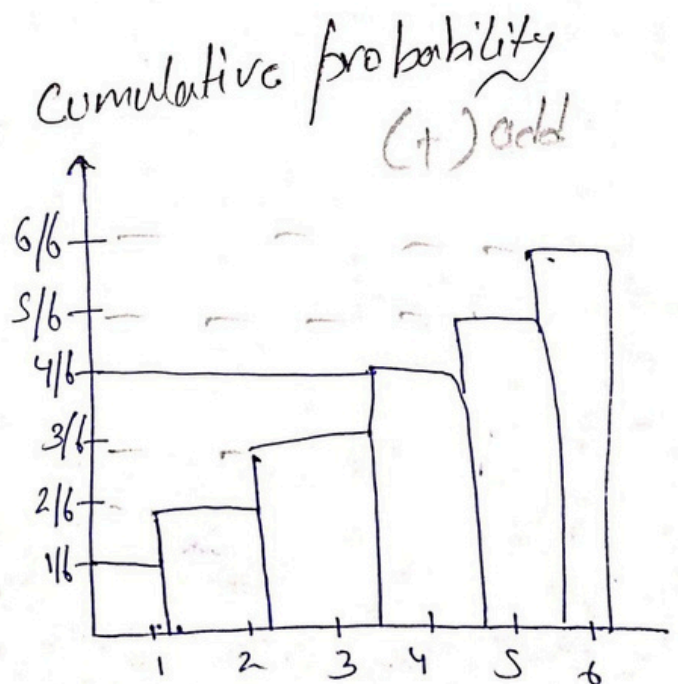
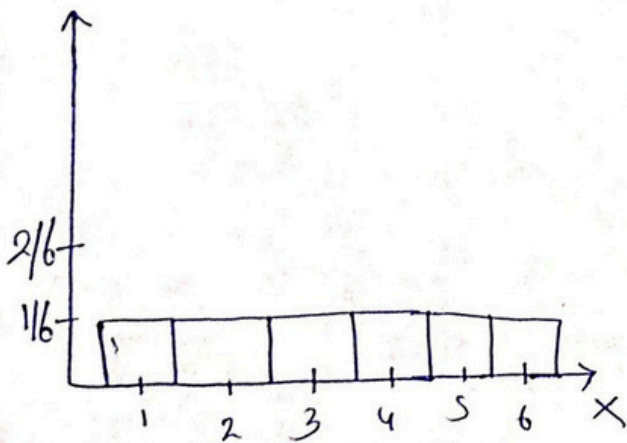
* Distribution of Continuous Random Variable



* Probability Mass function

* Discrete Random Variable.

Eg: Rolling Dice



Types of Probability Distribution:

- (i) Normal / Gaussian Distribution (pdf) $\rightarrow$ Continuous
- (ii) Bernoulli Distribution (pmf) $\rightarrow$ Discrete / binary
- (iii) Uniform Distribution (pmf)
- (iv) Poisson Distribution (pmf)
- (v) Log Normal Distribution (pdf)
- (vi) Binomial Distribution (pmf)

* Bernoulli Distribution:

* Discrete Random Variable (pmf)

* Outcomes are Binary (0, 1)

* $k = \{0, 1\}$

eg: Tossing a coin (H, T)

$$\begin{aligned} P_r(H) &= 0.5 = p \\ P_r(T) &= 0.5 = 1-p = q \end{aligned}$$

Let, whether the person will pass/fail

$$P_r(\text{Pass}) = 0.7 = p$$

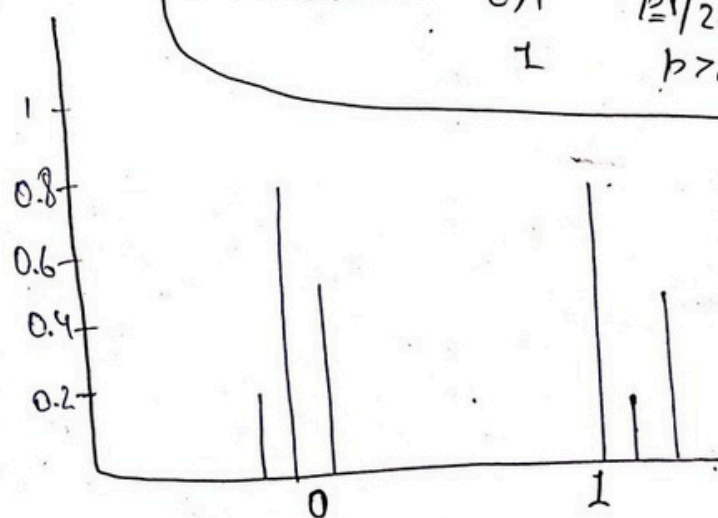
$$P_r(\text{fail}) = 1 - 0.7 = 0.3$$

$$0.3 = q$$

$$1-p = q$$

$$PMF = p^k \times (1-p)^{1-k} \quad \text{for } k = \{0, 1\}$$

$$\begin{aligned} * \text{ Variance} &= pq \\ * \text{ Std} &= \sqrt{pq} \\ * \text{ mean} &= p \\ * \text{ Median} &= \begin{cases} 0 & \text{if } p < 1/2 \\ 0.5 & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases} \end{aligned}$$



Three examples of Bernoulli Distributions

$$P(X=0) = 0.2 \text{ and } P(X=1) = 0.8$$

$$P(X=0) = 0.8 \text{ and } P(X=1) = 0.2$$

$$P(X=0) = 0.5 \text{ and } P(X=1) = 0.5$$

* Binomial Distribution: (pmf), $\{B(n, p), n = \{1, \dots, n\}\}$

* A single success/failure experiment is also called a Bernoulli trial and sequence of outcomes is said to be Binomial

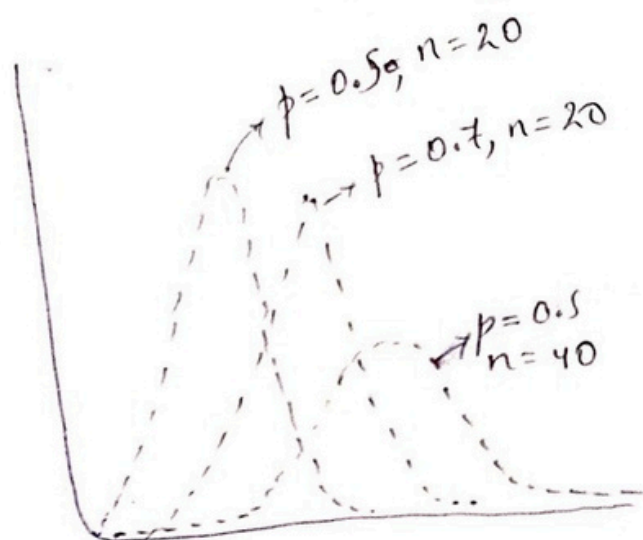
* Sequence of independent Bernoulli trial can collectively be Binomial distribution.

* Discrete Random Variable

* Every experiment outcome is binary.

* These experiment/trial is performed for n times

eg! Tossing a coin 10 times.



* Support! $K \in \{0, 1, 2, 3, \dots, n\} \rightarrow$ Number of success

PMF: $Pr(K, n, p) = {}^n C_K p^K (1-p)^{n-K}$

Mean $\rightarrow np$
Variance $\rightarrow npq$
std $\rightarrow \sqrt{npq}$

* Poisson Distribution: (pmf)

* Discrete Random Variable

* Describe the number of event occurring in a fixed time interval.

eg! No. of people visiting hospital every hour.



* $\lambda \rightarrow$ Expected No. of events to occur at every time interval.

* $T =$ Time interval

PMF = $\frac{e^{-\lambda} \lambda^x}{x!}$

$P(X=5) = \frac{e^{-3} 3^5}{5!}$
 $= 0.101$
 $= 10.1\%$

mean = $\lambda \times t$

variance = $\lambda \times T$
 $= \lambda \times t$

* Normal / Gaussian distribution (pdf)

* Continuous Random Variable

$$P.D.F = \left[\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right]$$

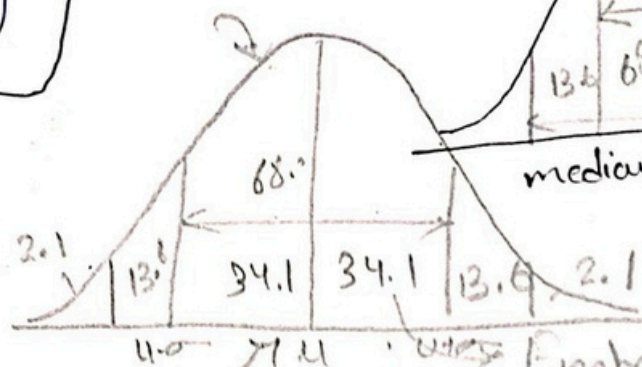
$$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2}$$

* mean = μ = Average

* Variance = σ^2

* Std = σ

Bell curve.



(68-95-97%)

Bell curve.

median = mode = mean

Empirical rule of Normal Distribution
 $\Rightarrow$ (68-95-97)%

eg: Weight of the student in the class
 Height " " " " " "

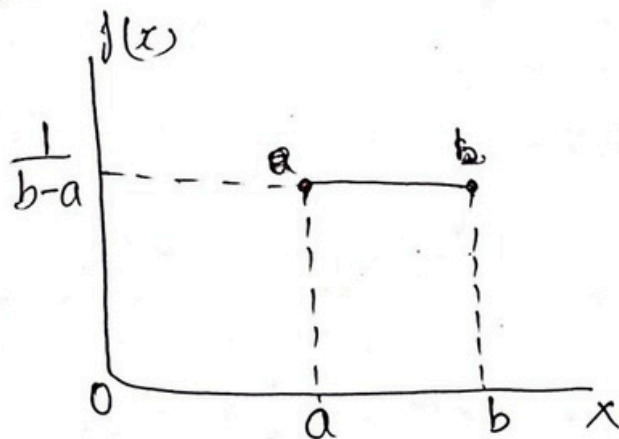
* Uniform Distribution

- * Continuous Uniform Distribution (pdf) $\rightarrow$ Rect
- * Discrete Uniform Distribution (pmf)

Continuous Uniform Distribution (pdf)

- * Continuous Random Variable
- * minimum & maximum values (a, b)
- * $-\infty < a < b < \infty$

$$pdf = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

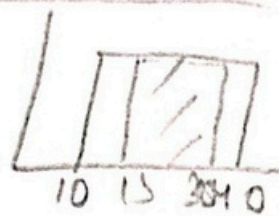


$$\text{Mean} = \frac{1}{2}(a+b)$$

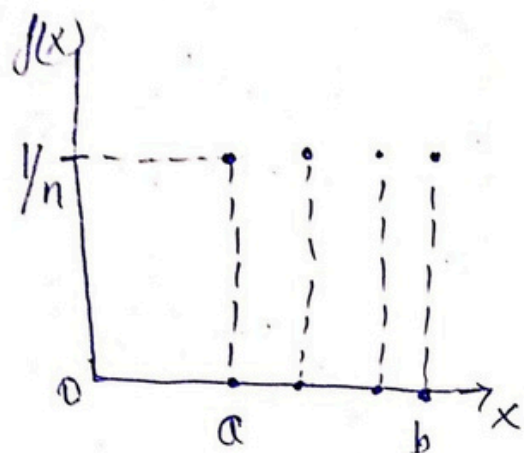
$$\text{Median} = \frac{1}{2}(a+b)$$

$$\text{Variance} = \frac{1}{12}(b-a)^2$$

Q1 = max 40 & min 10
 (i) fall between 15 & 30
 $P(15 \leq x \leq 30) = (x_2 - x_1) \times \frac{1}{b-a}$
 $= 15 \times \frac{1}{30} \Rightarrow 50\%$



* Discrete Uniform Distribution



* finite number of outcomes equally likely to happen.

* Discrete Uniform Distribution

eg: Rolling a dice $\rightarrow \{1, 2, 3, 4, 5, 6\}$

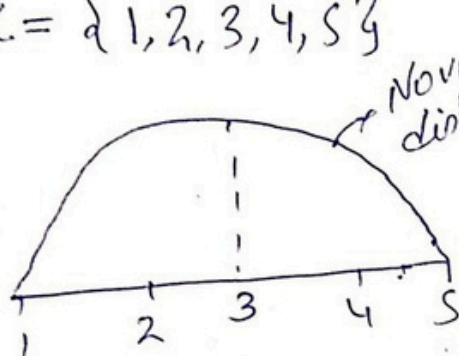
Notation $\rightarrow U(a, b)$

PMF $\rightarrow 1/n$

$$\text{mean, median} = \frac{a+b}{2}$$

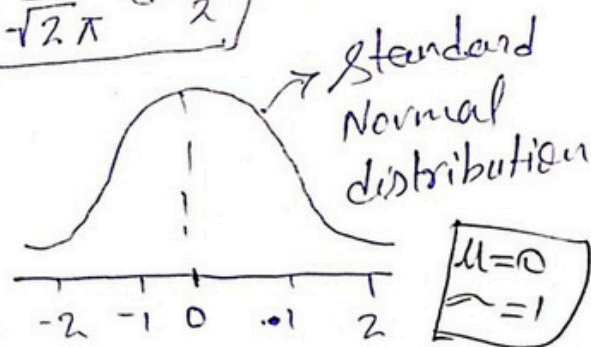
Standard Normal Distribution and Z-score

$X = \{1, 2, 3, 4, 5\}$



$\mu = 0$
 $\Rightarrow$
 $\sigma = 1$

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

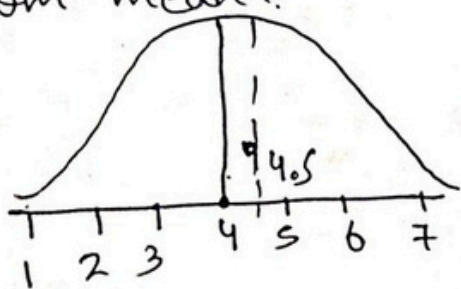


$$\mu = 0$$

$$\sigma = 1$$

* To Convert Normal distribution to a standard Normal distribution we use Z-score. $= \frac{x_i - \mu}{\sigma}$

Q1: How many student de stand- and deviation 4.5 is always from mean?



$$x_i = 4.5$$

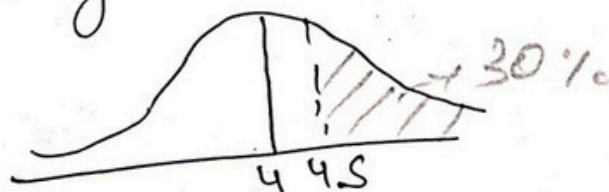
$$\mu = 4$$

$$\sigma = 1$$

$$So, = \frac{4.5 - 4}{1} = 0.5$$

Using Z score table

Q2: what percentage of data is falling above 4.5



$$So, Z score = \frac{4.5 - 4}{1} = 0.5$$

using Z score table

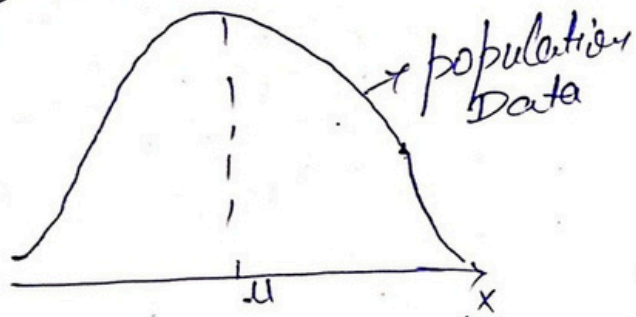
$$= 1 - 0.69146$$

$$= 0.3085$$

$$= 30\%$$

Central Limit Theorem (CLT) + Always Normal distribution

① $X \sim N(\mu, \sigma)$



taking

sample (n)

$$S_1 = \{x_1, x_2, x_3, \dots\} = \bar{x}_1$$

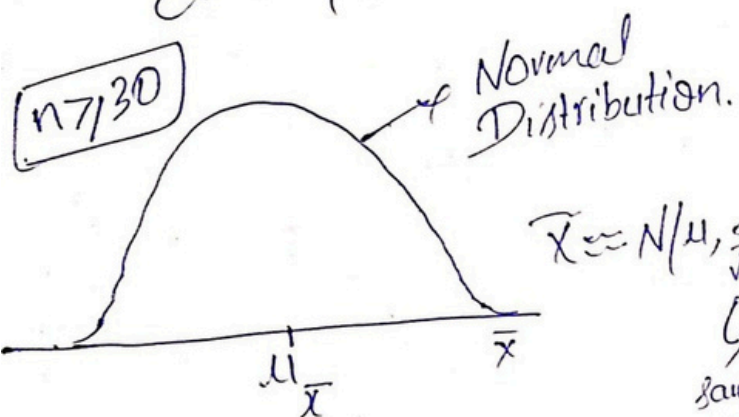
$$S_2 = \{x_1, x_3, \dots\} = \bar{x}_2$$

$$S_3$$

$$\vdots$$

$$S_{30} = \{ \dots \} = \bar{x}_n$$

So, By plotting that mean we get Normal distribution



Note

* Population if Normal distribute then N. of sample can be any

* If population is not normal distributed then N. of sample $n > 30$

$$\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

sample error of the mean

Estimate:

It is an observed numerical value used to estimate an unknown population parameter.

① Point Estimate

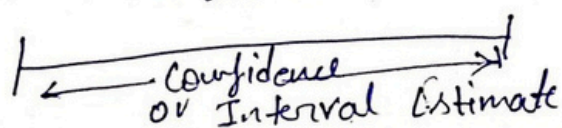
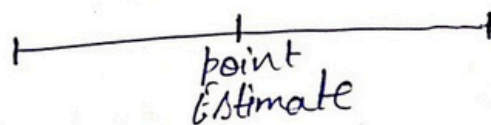
Single numerical value used to estimate the unknown parameters

* Sample mean is a point estimate of a population mean.

② Interval Estimate.

Range of values used to estimate the unknown population parameters

* Interval estimate of population parameters are called Confidence Intervals



$$C.I = \bar{x} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

critical value

sample mean

population

sample size

P value

The p-value is a number, calculated from a statistical test, that describes, how likely you are to have found a particular set of observations if the null hypothesis were true. P values are used in hypothesis testing to help decide whether to reject the null hypothesis

* Low p-value (< 0.05): strong evidence against the null hypothesis

→ Reject the null hypothesis

* High p-value (> 0.05): weak evidence against the null hypothesis

→ fail to reject the null hypothesis

Significance value:

Significance value is also known as alpha (α) (0.05, 0.01, 0.10)
Sr., 1%, 10%

Relationship b/w p-value & S.V

* If p-value $\leq \alpha$, reject the null hypothesis

* If p-value $> \alpha$, fail to reject null hypothesis

Confidence Level:

the probability that the Confidence Interval contains the true population parameter.

$$C.L = (1 - \alpha)$$

if: Significance value (α) = 0.05 then C.I = $1 - 0.05$
= 0.95

Error

Type 1 Error: We reject the Null Hypothesis when in reality it is true

Type 2 Error: We retain the null hypothesis when in reality it is false.

Hypothesis Testing and Statistical Analysis

- ① Z test $\rightarrow$ } Average
 ② t test $\rightarrow$ }
 ③ Chi Square $\rightarrow$ Categorical data
 ④ ANNOVA $\rightarrow$ Variance
- Z-Test is a statistical test used to determine whether there is a significant difference between sample and population mean or between two sample mean
- Z-Test
- * population std σ * $n \geq 30$
 - * data normally distributed
 - * Sample should be independent
- $Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ $\rightarrow$ population mean
- $\bar{x}$ sample mean

Q1

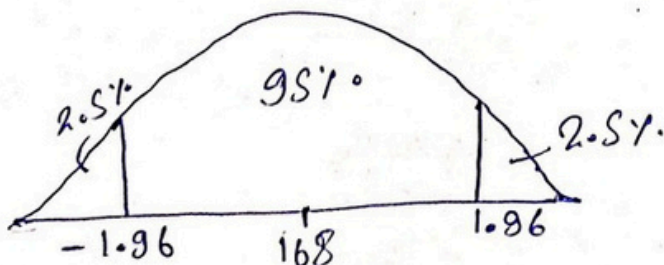
The average heights of all residents in a city is 168cm. with a $\sigma = 3.9$. A doctor believes the mean to be different. He measured the height of 36 individuals and found the average heights to be 169.5 cm.

- (a) state null and alternate hypothesis
 (b) At a 95% confidence level, is there enough evidence to reject the null hypothesis.

Sol: $\mu = 168$ $\sigma = 3.9$, $\bar{x} = 169.5$, $n = 36$

- a) Null Hypothesis H_0 $\mu = 168$ cm
 b) Alternate hypothesis H_1 $\mu \neq 168$ cm & 2 Tail Test

c) C.I = 0.95 $\alpha = 1 - 0.95 = 0.05$



Note! 2 tailed

- * Z test
- * p value

area under the curve = $1 - 0.05 = 0.9750$
 Using Z-score table we find (0.9750)
 $Z\text{-score} = 1.96$

d) Statistical Analysis / formula apply

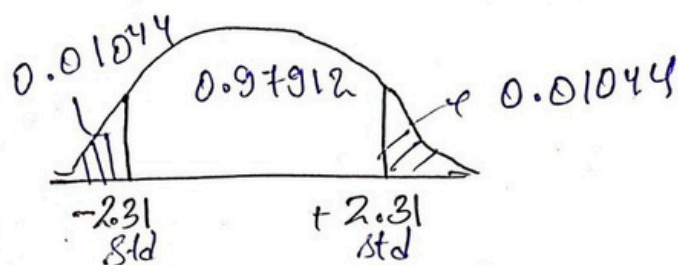
$$\boxed{Z\text{-score} = \frac{x_i - \mu}{\sigma}} \text{ population}$$

$$Z\text{-test} = \frac{x_i - \mu}{\sigma / \sqrt{n}} \Rightarrow \frac{169.5 - 168}{3.9 / \sqrt{36}} \Rightarrow 2.31$$

If Z-test value is less than -1.96 or greater +1.96 then, we Reject the Null hypothesis
else, we Accept Null Hypothesis

$\boxed{2.31 > +1.96}$ & we Reject the Null Hypothesis?

② P-value (By using)



$$Z\text{-test} = \frac{169.5 - 168}{3.9 / \sqrt{36}} \Rightarrow 2.31$$

$$= 1 - \text{area under the curve of } +2.31 \Rightarrow 1 - 0.98956 \Rightarrow 0.01044$$

$$p\text{value} = 0.01044 + 0.01044 = \underline{\underline{0.02088}}$$

if p-value < significance

$0.02088 < 0.05$ & we Reject the Null Hypothesis?

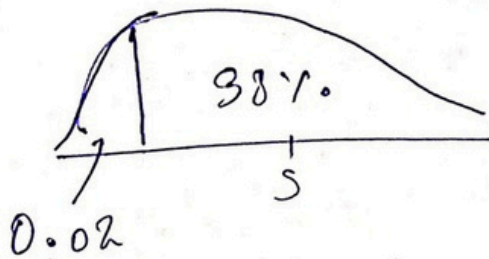
② A factory manufactures bulbs with a average warranty of 5 years with standard deviation of 0.50. A worker believes that the bulb will malfunctions in less than 5 years. He tests a sample of 40 bulbs and find the average time to be 4.8 years.

a) state null and alternate hypothesis

b) At a 2% significance level, is there enough evidence to support idea that the warranty should be revised?

Sol. $\mu = 5$, $\sigma = 0.50$, $N = 40$, $\bar{x} = 4.8$

- 1) Null Hypothesis $\Rightarrow H_0: \mu = 5$
 - 2) Alternative Hypothesis $H_1: \mu < 5$ & 1 tail Test?
 - 3) Decision Boundary
- $C.I = 0.98$
 $\alpha = 1 - 0.98 = 0.02$
 or $C.I = 1 - 0.02 = 0.98$



using z score of -2.53 we get 0.00570

4) P value $\rightarrow$ z-Test = $\frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$
 $= \frac{4.8 - 5}{0.50 / \sqrt{40}}$
 $= -2.53$



Area under the curve of -2.53, z-value is = 0.00570

p-value $\rightarrow$ 0.00570

$\therefore$ if p-value < significance
 $0.00570 < 0.02 \Rightarrow$ false $\therefore$ we accept the Null Hypothesis

T-test	z-test
Small ($n \leq 30$)	Larger ($n > 30$)
population variance unknown	known
Distribution used Student's t-distribution	Normal distribution

Do you know the population std - ?

Yes $\rightarrow$ Is the sample size above 30?

Yes $\rightarrow$ Z test

No $\rightarrow$ t test

No $\rightarrow$ t test

T-test

A t-test is a statistical test used to compare the mean of one or more groups when the population standard deviation is unknown and the sample size is small ($n < 30$).

* Assumptions:

- * The data should be numerical, not categorical.
- * The sample must be randomly selected: Random ^{sampling}
- * The data should follow an approximately normal distribution.

Formula

(a) one-sample

$$t = \frac{\bar{x} - \mu}{S / \sqrt{n}}$$

S = sample std

μ = population mean

$\bar{x}$ = sample mean

n = population size

(b) Two Sample

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$\bar{x}_1, \bar{x}_2$ = sample mean

S_1^2, S_2^2 = sample variance

n_1, n_2 = sample size

(c) paired T-test

$$t = \frac{\bar{d}}{\frac{S_d}{\sqrt{n}}}$$

$\bar{d}$ = Mean of difference b/w paired values

S_d = std of difference

n = Number of pairs

Degrees of freedom (df)

The Degree of freedom (df) represent the number of independent values in a statistical calculation that are free to vary.

(a) one sample

$$df = n - 1$$

n = sample size

(b) Two Sample

$$df = n_1 + n_2 - 2$$

n_1, n_2 = sample sizes of two group

(c) paired T-test

$$df = n - 1$$

n = Number of paired observations.

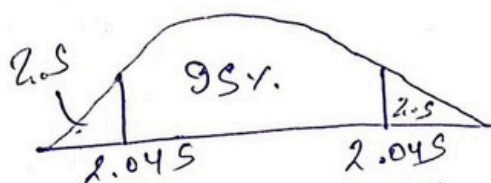
- * df determine the shape of t-distribution
- * smaller df mean wider distribution.
- * larger closer to normal

Q. t. Test.

In the population the average IQ is 100. A team of researchers want to test a new medication to see if it has either a positive or negative effect on intelligence or no effect at all. A sample of 30 participants who have taken the medication has a mean of 140 with a standard deviation of 20. Did the medication affect intelligence? C.I = 95%.

Sol: $\mu = 100$, $n = 30$, $S = 20$, $\bar{x} = 140$, C.I = 95%
 $\alpha = 0.05$

- ① Null Hypothesis $H_0: \mu = 100$
- ② Alternate Hypothesis $H_1: \mu \neq 100$ & 2 tail Test
- ③ $\alpha = 0.05$
- ④ Degree of freedom = $n - 1 = 30 - 1 = 29$
- ⑤ Decision Rule



By using t-table and the d.f in 29 for (2 tail) = 2.045
t-table 29 and (2 Tail) 0.05
= 2.045

If t test is less than -2.045 and greater than 2.045, then Reject the Null Hypothesis

- ⑥ calculate t test stat

$$t = \frac{\bar{x} - \mu}{S / \sqrt{n}} = \frac{140 - 100}{20 / \sqrt{30}} = 10.96$$

6) conclusion

we reject the null hypothesis

$$2.04 < 10.96$$

Confidence Interval and Margin of Error



We construct a confidence interval to help estimate what the actual value of the unknown population mean is

$$\text{Point Estimate} \pm \text{Margin of Error}$$

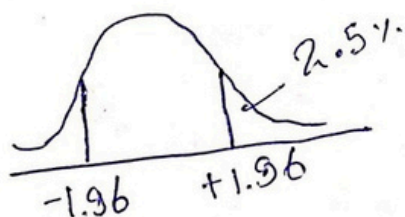
$$Z \text{ test} = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Q. In the verbal section of CAT exam the standard deviation is known to be 100. A sample of 25 test taken has a mean of 520. Construct a 95% C.I about the mean.

Sol

$$\bar{x} = 520, \sigma = 100, n = 25, \text{C.I} = 0.95, \alpha = 0.05$$

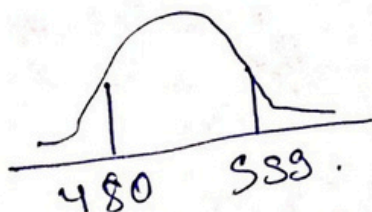
$$\begin{aligned} &= 1 - 0.05 \\ &= 0.9750 \text{ (using Z-table)} \\ &= 1.96 \end{aligned}$$



$$\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$\text{Lower C.I} = 520 - (1.96) \times \frac{100}{\sqrt{25}} = 480.8$$

$$\text{Higher C.I} = 520 + (1.96) \times \frac{100}{\sqrt{25}} = 559.2$$



So range between 480.8 to 559

So, I am 95% confident that the mean CAT score lies between 480.8 to 559.2

Chi-Square Test.

The Chi-Square Test help us check if two things are related or if the difference we see are just luck.

eg/ you ask 100 people if they like pizza or burgers, and you check if men and women have different preferences. The Chi-Square Test you if gender really affects their choice or if the difference happened by chance.

Types of Chi-Square Test:

(i) Chi square Goodness-of-fit Test

- * 1 categorical variable
- * Tests if a sample follow a known distribution
- * Use!: you compared observed vs expected frequencies

(ii) Chi-Square Test of independence

* 2 categorical variable.

- * Tests if two categorical variable are related or independent
- * you check if one category affects the others.

Formula! $\chi^2 = \sum \frac{(O - E)^2}{E}$

Q. In 2010 Census of the City the weight of the individuals in a small city were found to be the following.

< 50 kg	50-75	775
20%.	30%.	50

In 2020 ages of $n=500$ individuals were sampled. Below are result.

< 50	50-75	775
140	160	200

Using $\alpha=0.05$, would you conclude the population difference of weights has changed in the last 10 years.

Sol.

<50	50-75	775
20%	30%	50%

Expected

<50	50-75	775
140	160	200

Observed

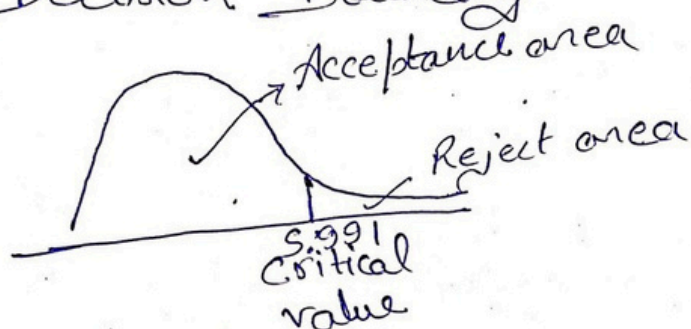
Expected

<50	50-75	775
$\frac{20 \times 500}{100}$	$\frac{30 \times 500}{100}$	$\frac{50 \times 500}{100}$
100	150	250

- Null Hypothesis: H_0 : The data meets the exception
- Alternate Hypothesis: H_1 : The data does not meet the exception.
- $\alpha = 0.05$ CI = 95%.

④ Degree of freedom $df = K - 1 = 3 - 1 = 2$

④ Decision Boundary



By using Chi-table
 $df = 2$ & $\alpha = 0.05$
 we get 5.991

⑤ calculate chi square test statistics

$$\chi^2 = \sum \frac{(O - E)^2}{E} \Rightarrow \frac{(40)^2}{100} + \frac{(10)^2}{150} + \frac{(50)^2}{250} = 26.66$$

⑥ Con^n
 If χ^2 is greater than 5.99; Reject H_0
 else we fail to reject the Null Hypothesis.

So, $\chi^2 = 26.66 > 5.99$ { Reject the Null Hypothesis }

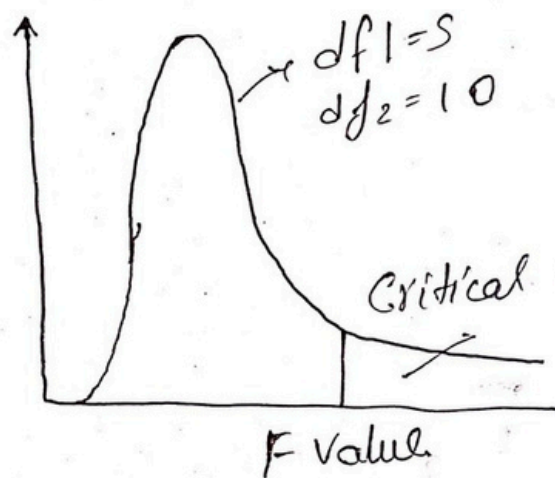
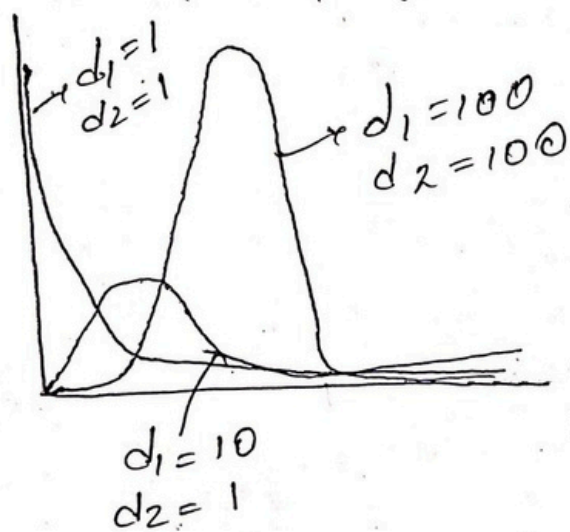
F-Distribution

The F-Distribution is a probability distribution used mainly in ANOVA (Analysis of Variance) and F-Test to compare of two datasets. It help determine if the mean variance the difference in data are real or just due to chance.

Key features of F-Distribution

- * Always positive (values are never negative)
- * Right-skewed (not symmetric like a normal distribution)
- * Used in hypothesis testing when comparing two variance
- * Defined by two degree of freedom (df_1, df_2) > 0 always

Probability density function



Formula

$$pdf = f(x; d_1, d_2) = \frac{\sqrt{\frac{(d_1 x)^{d_1} d_2^2}{(d_1 x + d_2)^{d_1 + d_2}}}}{x B\left(\frac{d_1}{2}, \frac{d_2}{2}\right)} \rightarrow \text{Beta function or } B(m, n)$$

$$B(m, n) = \frac{(m-1)!(n-1)!}{(m+n-1)!} = \frac{m+n}{mn} / \frac{(m+n)}{m}$$

$$F = \frac{S_1^2}{S_2^2} \dots S_1^2, S_2^2 \text{ variance of first and second sample.}$$

F-Test

The F-Test is a statistical test used to compare the variance of two datasets. It helps check if the spread (variability) of data in two groups is significantly different.

Q The following data shows the no. of bulbs produced daily for some days by 2 workers A and B

A

40	30	38	41	38	35
----	----	----	----	----	----

B

39	38	41	33	32	39	40	34
----	----	----	----	----	----	----	----

Can we consider based on the data workers B is more stable and efficient. $\alpha = 0.05$ C.I = 95% = 0.95

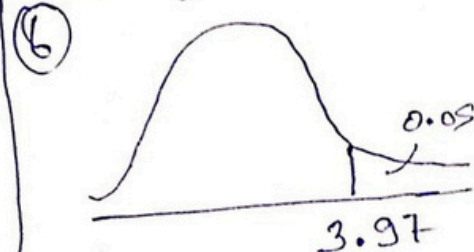
Sol: ① Null Hypothesis $H_0: \sigma_1^2 = \sigma_2^2$
② Alternate Hypothesis $H_1: \sigma_1^2 \neq \sigma_2^2$

③ calculate of variance

A	x_i	$(x_i - \bar{x})^2$
	40	37
	30	37
	38	37
	41	37
	38	37
	35	37
$\bar{x}_1 = 37$		$\sum (x_i - \bar{x})^2 = 80$

B	x_i	$(x_i - \bar{x})^2$
	39	37
	38	37
	41	37
	33	37
	32	37
	39	37
	40	37
	34	37
$\bar{x}_2 = 37$		$\sum (x_i - \bar{x})^2 = 84$

By using F-table and $df_1 = 5$ and $df_2 = 7$



④ So, if F test is greater than 3.97; Reject the Null Hypothesis
 $1.33 < 3.97$
We fail to reject Null

$$s_1^2 = \frac{\sum_{i=1}^n (x_i - \bar{x}_1)^2}{n-1}$$

$$= \frac{80}{6-1}$$

$$= 16$$

$$s_2^2 = \frac{84}{8-1}$$

$$= \frac{84}{7}$$

$$= 12$$

⑤ Calculate of variance ratio of F-test

$$F = \frac{s_1^2}{s_2^2} = \frac{16}{12} = 1.33$$

⑥ Decision Rule

$$df_1 = 6-1 = 5 \text{ and } df_2 = 8-1 = 7$$

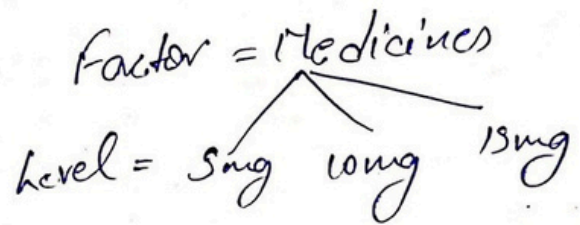
ANOVA (Analysis of Variance)

ANOVA is a statistical method used to compare the mean of 2 or more groups

Anova

- ① Factors (Variables)
- ② Level

eg!



Assumption in ANOVA

- (i) Data is numerical and normally distributed
- (ii) Homogeneity of Variance
Each one of the population has same variance
 $[\sigma_1^2 = \sigma_2^2 = \sigma_3^2]$
population variance in different levels of each independent variable are equal.
- (iii) Samples are independent and random
- (iv) Absence of outliers.

formula

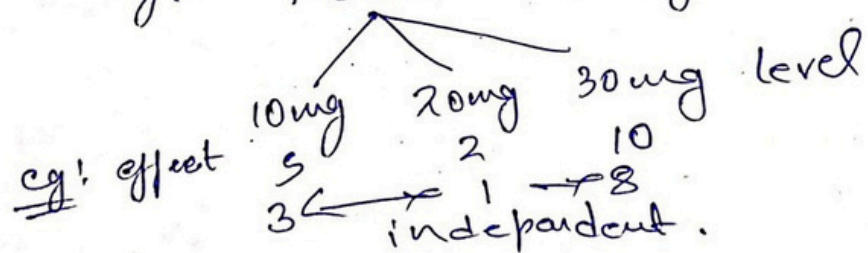
$$F = \frac{\text{Variance between groups}}{\text{Variance within groups}}$$

Types of ANOVA

(i) One way ANOVA

1 factor with atleast 2 levels those level are independent.

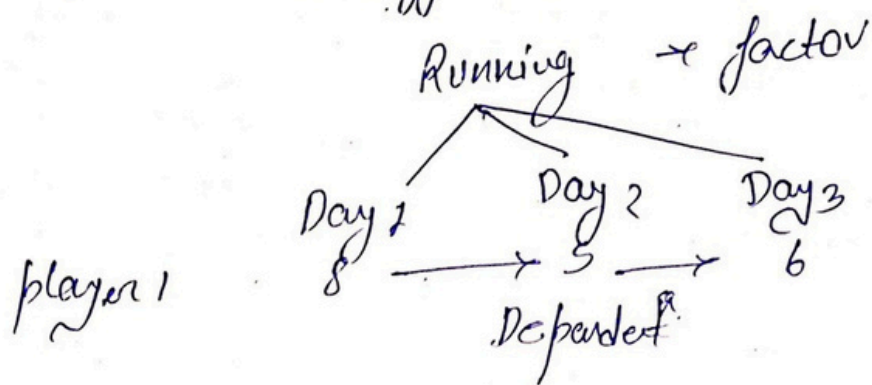
eg! Medication $\rightarrow$ factor



② Repeated Measures ANOVA:

One factor with atleast 2 levels, ^{and level} are dependent.

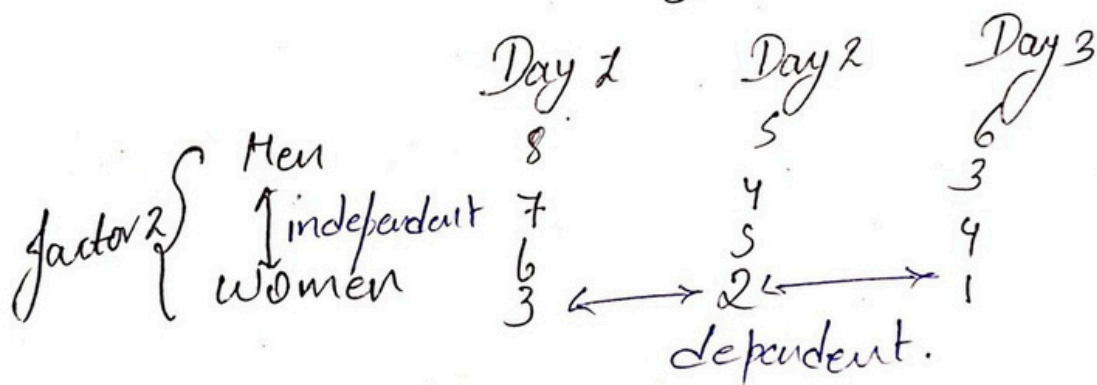
* It used when you measure the same subjects multiples times under different conditions or over time.



③ Factorial Anova / Two factor

Two factor or more factor each of which with atleast 2 level, level can be either independent and dependent

Running - Factor 1



Hypothesis Testing in Anova

Null Hypothesis H_0 : $\mu_1 = \mu_2 = \mu_3 = \dots = \mu_R$

Alternate hypothesis H_1 : Atleast one of the mean is not equal.

$$\times \boxed{\mu_1 \neq \mu_2 = \mu_3 = \dots = \mu_R}$$

Q.1 Doctors want to test a new medication with reduce headache. They split the participant into 3 condition [15mg, 30mg, 45mg]. Later on the doctor ask the patient to rate the headache between [1-10]. Are they any differences between the 3 condition using alpha 0.05?

So! ① Null Hypothesis: $H_0 = \mu_{15} = \mu_{30} = \mu_{45}$
 ② Alternate Hypothesis H_1 : not all μ 's are equal.

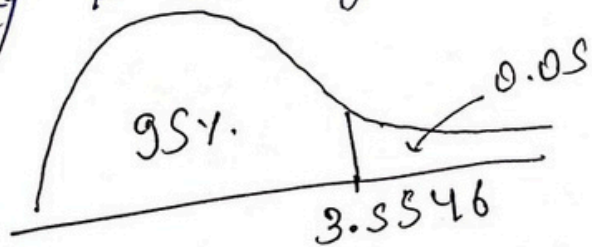
$\alpha = 0.05$, C.I = 0.95

15mg	30mg	45mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

③ calculate Degree of freedom
 $N = 21$ $a = 3$ $n = 7$

df between = $a - 1 = 3 - 1 = 2$
 df within = $N - a = 21 - 3 = 18$
 df total = $N - 1 = 21 - 1 = 20$ (df between + df within)

Using F-table of (2, 18) we get = 3.5546
 Critical = 3.5546



④ State Decision Rule

if F is greater than 3.5546, reject the Null Hypothesis

⑤ calculate test statistics

	SS	df	MS
Between	98.67	2	$\frac{98.67}{2} = 49.34$
within	10.29	18	$\frac{10.29}{18} = 0.54$
Total	108.95	20	49.88

$$F = \frac{49.34}{0.54} = 86.56$$

SS_{between}, SS_{within}, SS_{total}

$$SS_{\text{between}} = \frac{\sum (\sum a_i)^2}{n} - \frac{T^2}{N}$$

$$15\text{mg} = 9+8+7+8+8+9+8 = 57$$

$$30\text{mg} = 7+6+6+7+8+7+6 = 47$$

$$45\text{mg} = 4+3+2+3+4+3+2 = 21$$

15mg	30mg	45mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

$$SS_{\text{between}} = \frac{57^2 + 47^2 + 21^2}{7} - \frac{[57+47+21]^2}{21} = 98.67$$

$$SS_{\text{within}} = \sum y^2 - \frac{\sum (\sum a_i)^2}{n}$$

$$= \sum y^2 - \left[\frac{57^2 + 47^2 + 21^2}{7} \right]$$

$$\sum y^2 = 9^2 + 8^2 + 7^2 + 8^2 + 8^2 + 9^2 + 8^2 + 7^2 + 6^2 + \dots$$

$$= 853$$

$$= 853 - \left[\frac{57^2 + 47^2 + 21^2}{7} \right]$$

$$SS_{\text{within}} = 10.29$$

$$\textcircled{5} \quad \underline{SS_{\text{total}}} = \sum y^2 - \frac{T^2}{N}$$

$$= 853 - \frac{125^2}{21} = 108.95 \quad (SS_{\text{between}} + SS_{\text{within}})$$

$$\textcircled{6} \quad F = \frac{MS_{\text{between}}}{MS_{\text{within}}}$$

$$= \boxed{\frac{49.34}{0.84} = 86.56}$$

So, conⁿ

If F is greater than 3.5546, reject the Null Hypothesis

$$\boxed{86.56 > 3.5546} \quad \text{Reject the Null Hypothesis}$$